

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru-3bb728174 | github.com/zac-09 | Kampala, Uganda

PROFESSIONAL SUMMARY

Platform Engineer with 4+ years of experience in cloud infrastructure, site reliability engineering, and zero-downtime migrations. Led the end-to-end migration of 20+ production applications from Parse/MongoDB to Firebase/GCP with zero downtime, cutting \$5,000/month in infrastructure costs. Deep expertise in TypeScript, Python, GCP (Pub/Sub, Firestore, Cloud Functions), and AWS. Built real-time data sync pipelines, architected microservices with Kubernetes and Docker, and championed operational excellence through documentation and team training. Skilled at diagnosing complex distributed system problems and building reliable, scalable infrastructure.

CORE COMPETENCIES

Site Reliability Engineering

Zero-Downtime Migrations

Cloud Infrastructure (GCP/AWS)

Kubernetes & Docker

TypeScript & Python

CI/CD Pipelines

Observability & Monitoring

Incident Response

WORK EXPERIENCE

MTailor Sep 2022 – Present

Software Engineer / Platform Engineer — US Remote

- Led **zero-downtime migration** of 20+ production applications from Parse/MongoDB to Firebase/GCP, reporting directly to CTO
- Built **real-time two-way sync pipeline** between MongoDB and Firestore using Node.js and Google Pub/Sub for reliable message processing
- Implemented **monitoring and alerting** for the sync pipeline to ensure data integrity during parallel production traffic
- Migrated file storage from Amazon S3 to GCS via Python automation script with validation
- Reduced infrastructure costs by \$5,000/month** by consolidating services from AWS to GCP
- Trained Ops team on new Firebase Dashboard, backup procedures, and incident response runbooks
- Built 3D visualization feature using video overlay and ffmpeg library, increasing buyer conversion

MTailor Mar 2022 – Jul 2022

Contractor — US Remote

- Executed complex backend migration from Parse/MongoDB to Firebase Firestore and GCP
- Built migration script to read data from MongoDB and write to Firestore with complex processing logic
- Wrote technical documentation for engineers on working with new Firebase SDKs (Node.js and Python)

CodeBits Jan 2020 – Jul 2021

Software Engineer — Uganda Remote

- Led team of 4 developers building legal tech systems for NGOs in Uganda
- Architected microservices backend** for case management using Apache Kafka, Docker, and **Kubernetes**
- Built containerized infrastructure with Docker for reliable deployment and scaling
- Developed cross-platform mobile app using React Native with Firebase push notifications (iOS & Android)
- Built USSD service communicating with Node.js backend via gRPC for low-connectivity access

Dr Wealth Aug 2021 – Jan 2022

Software Engineer — Singapore Remote

- Optimized PWA delivering real-time stock market prices from Firebase backend
- Built ad-hoc jobs to query over 2 million Firestore records for price synchronization
- Extended backend APIs using Firebase Cloud Functions and Express.js

PROJECTS

Real-time Pub/Sub Sync Pipeline Infrastructure

Built live MongoDB ↔ Firestore two-way sync supporting parallel production traffic during zero-downtime migration. Implemented monitoring, message deduplication, and rollback capabilities.

Node.js, Google Pub/Sub, MongoDB, Firestore, GCP

Microservices Case Management System Kubernetes

Architected distributed backend for FIDA Uganda using event-driven microservices. Handled legal case tracking for NGO operations across Uganda.

Apache Kafka, Docker, Kubernetes, Node.js, gRPC

Healthcare WhatsApp Chatbot AI Integration

Integrated OpenAI + Google Vision for prescription OCR via WhatsApp Business API. Enabled healthcare access through messaging platform.

Node.js, OpenAI API, Google Vision, WhatsApp Business API

EDUCATION

Bachelor of Science in Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go (learning) **Cloud:** GCP (Pub/Sub, Firestore, Cloud Functions), AWS (S3, EC2, EBS)

Infrastructure: Kubernetes, Docker, CI/CD, Linux **Databases:** MongoDB, Firestore, SQL

Frameworks: Node.js, Express.js, React, React Native **Tools:** Git, Firebase, Apache Kafka, gRPC